

Explainer on the Interconnection Final Rule

Improvements to Generator Interconnection Procedures and Agreements, Docket No. RM22-14-000, Order No. 2023

Please note that on March 21, 2024, the [Commission issued Order 2023-A](#) that revises or clarifies parts of the rule described in the explainer below. OPP published an [explainer on Order 2023-A](#).

Introduction

On July 28, 2023, the Federal Energy Regulatory Commission (FERC or Commission) issued a new rule to reform procedures and agreements that electric transmission providers use to integrate new generating facilities into the existing transmission system, sometimes referred to as the “electric grid” or “grid.”^[1] Designated as Order No. 2023, FERC adopted these reforms to reduce backlogs for projects seeking to connect to the transmission system, improve certainty in the interconnection processes managed by the dozens of transmission providers around the country, and ensure access to the transmission system for new technologies.

The final rule will take effect on **November 6, 2023**. Originally, transmission providers had until December 5, 2023 to file compliance filings with FERC. However, in response to requests for extension, FERC extended the compliance deadline to **April 3, 2024**. On **March 21, 2024**, FERC issued a rehearing order (Order 2023-A) that made certain minor changes to Order No. 2023, but did not change the overall goal of reforming and streamlining grid interconnection processes.^[2] FERC extended the compliance deadline for Order 2023-A to 30 days after publication of the Order in the *Federal Register*, or May 16, 2024.

In these compliance filings, transmission providers, including regional transmission organizations (RTOs) and independent system operators (ISOs),^[3] will propose how they would implement the requirements outlined in the final rule. Once the Commission issues orders approving these compliance filings, the transmission providers will be required to implement the final rule’s new requirements.

This explainer offers a brief summary of the final rule and should not be relied on as a legal document. Additional details and the final regulatory text that governs these reforms is available at [Order No. 2023](#).

Interconnection Reform: From Notice of Proposed Rulemaking to Final Rule

Q. How did the Commission decide on the reforms in this order?

A. A final rule sets out new or revised requirements, or removes existing requirements, for regulated entities that must comply with the requirements. A final rule also states when the requirements will become effective, with actual implementation to follow FERC’s approval of the regulated entity’s compliance filing. When preceded by a Notice of Proposed Rulemaking (NOPR), a final rule will identify significant substantive issues raised by commenters in response to the NOPR and explain the agency’s responses.^[4]

In June 2022, FERC issued a [NOPR](#) proposing several changes to existing generator interconnection processes and invited comments from the public. After the public comment period closed, the Commission reviewed and analyzed all comments received and addressed them in the final rule. The final rule adopted many of the processes and reforms proposed in the NOPR, while revising other reforms and declining to adopt others based on the record developed through public comment.^[5]

The NOPR asked, among other things, whether there was a need to reform the interconnection process in the first place. The vast majority of commenters agreed with the Commission that there was a need for reform. Many of these commenters underlined the large number of generating facilities attempting to interconnect to the

transmission system (in lines referred to as “interconnection queues”) which have resulted in severe interconnection delays and backlogs across the country.[6]

Supporting this view, recent data shows that the combination of rising numbers of interconnection requests and limited transmission capacity are increasing interconnection queue backlogs.[7] As of the end of 2022, there were over 10,000 active interconnection requests in interconnection queues throughout the United States, representing over 2,000 gigawatts (GW) of potential generation and storage capacity. Delays in the interconnection study process have also contributed to interconnection queue backlogs.[8] For instance, of the 2,179 interconnection studies completed in 2022, 68% were issued late.[9]

Q. What happens next?

A. As noted above, the final rule becomes effective on November 6, 2023. After publication, transmission providers have until April 3, 2024, to propose how they will operate in compliance with the final rule by proposing changes to their tariffs. Transmission providers propose these changes in what is known as a “compliance filing” that they submit to the Commission for review.

Each transmission provider will need to demonstrate its compliance with the requirements of the final rule in one of two ways. On the one hand, the transmission provider can adopt, without changes, the tariff language set out in the final rule. On the other hand, the transmission provider can propose deviations from the requirements adopted in this final rule and demonstrate how those deviations satisfy the final rule’s requirements. FERC will consider these deviations on a case-by-case basis.

Transmission providers proposing deviations from the final rule will be held to either the “independent entity variation” standard (for RTOs/ISOs) or the “consistent with or superior to” standard (for non-RTO/ISO providers). [10]

If you have questions regarding the final rule, or need assistance accessing the Commission’s administrative record or web resources, please contact OPP by e-mail at OPP@ferc.gov or by phone at (202)502-6595.

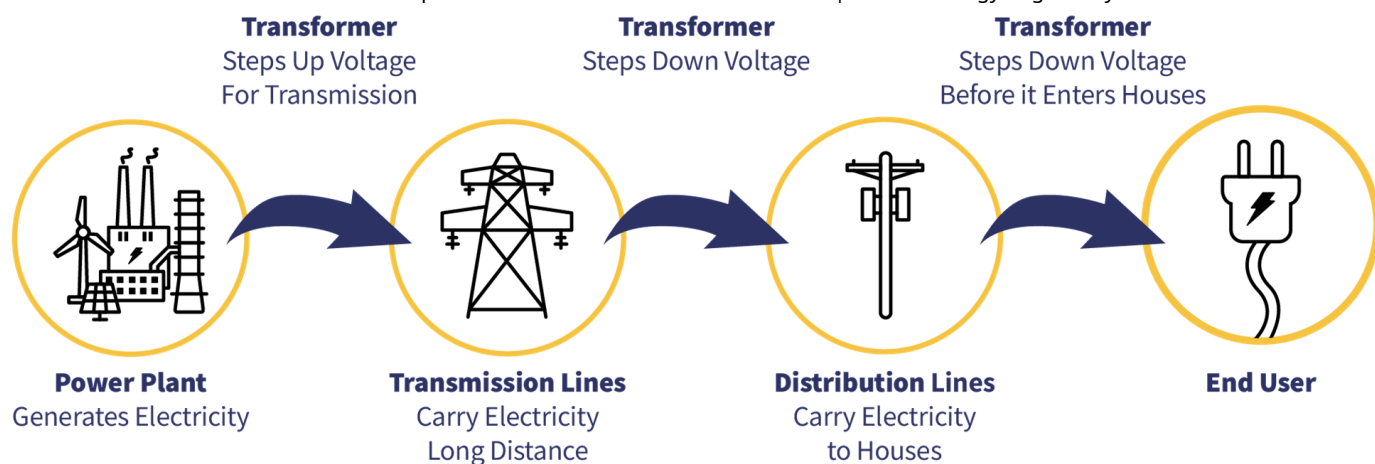
The Interconnection Final Rule

Q. What does this final rule address?

A. The final rule lays out how FERC will reform processes used by transmission providers to study and connect generating facilities to the transmission system.

Traditionally, transmission providers studied requests to interconnect generating facilities one by one as they came in, sometimes referred to as a “first-come, first-served” serial process. This process worked well when transmission providers were processing a smaller amount of projects each year, many of which were hydropower, nuclear, coal, and natural gas resources and often large in size. Transmission providers now face a significant increase in requests to interconnect, including wind, solar, and battery storage projects. This increase has challenged the interconnection process, making it take longer to interconnect new generating facilities to the transmission system that will ultimately carry electricity from those facilities to homes and businesses.

Generally, the final rule will require transmission providers to study projects in batches or “clusters,” such that the new approach is often called a “cluster study process.”[11] Other changes include penalties for transmission providers that fail to complete studies on time, stricter financial readiness and site control requirements for interconnection customers that are intended to discourage speculative projects, and measures that could make it easier to integrate generating facilities with unique operating characteristics, such as battery storage projects, into the transmission system. The final rule thus streamlines the interconnection process to ensure that new generation, especially generation representing advanced technologies, can connect to the transmission system in a timely manner.



FIRST KEY AREA: Transitioning from a First-Come, First-Served Serial Process to a First-Ready, First-Served Cluster Study Process

Transmission providers will study proposed generating facilities in batches or clusters, rather than separate studies for each individually. To ensure that only ready projects can enter and proceed through the queue, transmission providers also will impose additional financial readiness and site control requirements on interconnection customers.

Facilitating Public Interconnection Information Access

FERC identified the need to give prospective interconnection customers information that can help them make more informed decisions prior to entering the interconnection queue. To this end, the final rule requires transmission providers to maintain a publicly available visual representation, also known as a heatmap, of available transmission capacity. FERC intends for the heatmaps to benefit prospective interconnection customers by helping them to identify ideal points of interconnection based on areas of expected congestion.

Cluster Study Replaces Serial Study

As noted, the final rule directs transmission providers to group interconnection requests into clusters for study. Cluster studies are now the required interconnection study method, and all transmission providers will need to eliminate the serial study process and instead use a cluster study process.

- Timelines - The final rule includes a 150-day cluster study, followed by a facilities study before the interconnection customer enters into an interconnection agreement.
- Restudies - If a given cluster needs to be restudied (for example, if a major project with a higher or equivalent queue position drops out of the queue), the final rule provides for restudies.
- Allocation of Cluster Study Costs – The final rule requires transmission providers to establish a method to allocate the cost of the single, shared cluster study among the interconnection customers participating in the cluster study based on both the size of the group and the size of the projects.
- Allocation of Cluster Network Upgrade Costs – Beyond the study costs, the final rule addresses how the cluster process defines which interconnection customers pay and how much they pay for certain transmission upgrades called “network upgrades” needed to accommodate the new generation facilities being studied.
 - The final rule allocates network upgrade costs to interconnection customers within a cluster using a “proportional impact method,” which is a technical analysis that determines the degree to which each generating facility contributes to the need for a specific network upgrade. This cost allocation method thus shares network upgrade costs among several interconnection customers that may benefit from - and cause the need for - certain network upgrades.

Increased Financial Readiness and Site Control Requirements

The final rule’s overall aim is to ensure that interconnection queues include only projects likely to be built. A key need for reform is that interconnection practices have permitted an interconnection customer to proceed through the process without having to show sufficient evidence of ability to develop its project. Requiring greater financial readiness deposits and demonstration of site control is expected to curb speculative interconnection requests, reduce congestion in the queue, and enable viable projects to move more quickly through the process.

The four main ways that the final rule tackles this issue is by (1) increasing study deposit amounts, (2) requiring demonstration of site control, (3) requiring a commercial readiness deposit, and (4) imposing withdrawal penalties.

- Increasing Study Deposit Amounts - The final rule will require interconnection customers to pay the transmission provider a deposit for carrying out interconnection studies. Due at the time of the interconnection request, the study deposit amount will be based on the MW size of proposed generating facility.
- Demonstration of Site Control - Site control is generally defined as evidence that demonstrates that the interconnection customer has the right to develop the land or site where its generating facility will be built. This evidence may include documentation that the interconnection customer has the right to occupy and develop the site, such as proof of ownership or a lease.
 - Under the final rule, interconnection customers must provide evidence of 90% site control at the time of submission of the interconnection request, and 100% site control at the time of execution of the facilities study agreement.
 - Under limited circumstances, interconnection customers are allowed to use a deposit instead of site control demonstrations where regulatory limitations may prohibit customers from obtaining site control. For example, federal, state, Tribal, or local laws may, in some circumstances, make it infeasible for interconnection customers to otherwise demonstrate site control in a timely manner.
- Commercial Readiness - The final rule requires interconnection customers to submit a “commercial readiness” deposit at the beginning of each study in the cluster study process. At later stages of interconnection study, the commercial readiness deposit framework shifts from one based on generating facility size to one based on increasing percentages of the interconnection customer’s identified network upgrade costs.
- Withdrawal Penalties - The final rule requires the transmission provider to impose a withdrawal penalty if the withdrawal of an interconnection customer from the queue has a material impact on the cost or timing of any interconnection requests with an equal or lower queue position. These withdrawal penalties are meant to deter non-viable projects from entering or remaining in the interconnection queue and to mitigate potential harm to other interconnection customers in the queue.
- Transition Process - FERC recognized that the final rule could represent a big change in how transmission providers process interconnection requests. To ease the switch to the new requirements, the final rule requires transmission providers to offer interconnection customers three options depending on the progress of the interconnection request. During the transition, existing interconnection customers in the queue may select a transitional serial facilities study, select a transitional cluster study, or withdraw from the interconnection queue without penalty.

SECOND KEY AREA: Increase the Speed of Interconnection Queue Processing

The final rule imposes firm deadlines and penalties if transmission providers fail to complete interconnection studies on time (with certain exceptions). Additionally, the final rule requires a detailed process for studying the impacts of proposed generating facilities on neighboring transmission systems (called “affected systems”), including a specific modeling standard and *pro forma* affected system agreements.

- Elimination of the Reasonable Efforts Standard – Prior to the final rule, transmission providers were only required to make “reasonable efforts” to meet study deadlines, and there were little to no consequences if transmission providers missed deadlines. FERC eliminated the “reasonable efforts” standard and introduced penalties if a transmission provider fails to meet study deadlines.
- Affected Systems – Interconnection customers whose requests have reliability impacts on affected systems, such as neighboring transmission providers, may be responsible for additional network upgrade costs to mitigate those impacts. Yet, affected system studies that analyze these impacts were previously conducted inconsistently. This has challenged interconnection processing by leading to late-stage project withdrawals when affected system network upgrades are identified late in the process.
 - To ensure timely affected system studies, the final rule sets firm deadlines for the transmission provider to provide initial notice to the affected system and for the affected system to state its intention to

conduct an affected system study. The affected system must then conduct a 150-day cluster study and use the proportional impact method to allocate among interconnection customers any network upgrade costs.

- The final rule further proposes standardized affected systems agreements for the affected system studies and the construction of any required network upgrades. These standardized agreements are meant to provide consistency, as well as greater transparency and more timely information concerning costs, during the interconnection process.

THIRD KEY AREA: Reforms to Incorporate Technological Advancements in the Interconnection Process

The final rule seeks to create efficiencies for resources able to locate on the same site. The final rule requires transmission providers to allow projects to co-locate on a shared site behind a single point of interconnection and share a single interconnection request, thereby removing barriers for co-located resources by creating a more efficient standardized procedure for these types of configurations.

The final rule also allows interconnection customers to add a generating facility to an existing interconnection request, as long the total amount of generation capacity in the original interconnection request is unchanged, without automatically losing their queue position because of changes to the original request. Further, the final rule requires transmission providers to consider alternative transmission solutions.

Additionally, recognizing the impacts of nonsynchronous generating facilities (e.g., wind, solar, and battery storage resources) on transmission conditions and operations, the final rule requires that transmission providers provide more granular and accurate modeling data about these resources and follow capability standards to ensure system reliability.

- Increasing Flexibility - Historically, interconnection requests have been limited to a single generating facility seeking to interconnect to the transmission system. However, given advancements in energy technologies – particularly storage technologies - the final rule requires transmission providers to allow more than one resource to co-locate on a shared site behind a single point of interconnection and share a single interconnection request. This clarifies the process for different types of generating facilities that may prove complementary, such as solar combined with storage, wind combined with solar, or natural gas combined with wind and storage.
- Incorporating Alternative Transmission Technologies – Because alternative transmission technologies can provide substantial benefits due to their quick deployment and lower costs compared to other types of network upgrades, the final rule requires transmission providers when conducting a cluster study to evaluate alternative transmission technologies, which may include advanced conductors, advanced power flow control, transmission switching, voltage source converters, static synchronous compensators, static VAR compensators, synchronous condensers, and tower lifting. [\[12\]](#)
- Modeling and Performance Requirements for Non-Synchronous Generating Facilities - Until roughly the 1990s, electricity in the United States was primarily generated from a mix of hydropower, nuclear, coal, and natural gas resources. The physical and mechanical properties of these “synchronous” generation resources, which inherently produce AC power, underpin key services supporting reliable operation of the transmission system. Further, synchronous resources are able to stay connected to the grid during disturbances, a capability known as “ride through”.

Over the years, the generation mix has changed with more electricity generated by non-synchronous resources, such as wind, solar photovoltaic, and battery storage. These non-synchronous resources do not inherently provide key grid services but can be programmed to do so. In addition, the “ride through” capability of non-synchronous resources must be programmed into the resource. Otherwise, these non-synchronous generating facilities may not react to disturbances on the transmission system in the same way as synchronous resources, and this sensitivity can make them less able to withstand system disturbances without tripping offline or reducing power – which, in the aggregate, may impact the reliability of the transmission system.

To this end, the final rule establishes a variety of requirements regarding modeling and data used in interconnection studies to, for example, more accurately model non-synchronous facilities. The final rule also

introduces certain capability requirements for those facilities once they are interconnected, in particular to ensure “ride through.”

[1] FERC's Office of Public Participation published an explainer that discusses the Notice of Proposed Rulemaking (NOPR) on interconnection and how the interconnection process works at www.ferc.gov/explainer-interconnection-notice-proposed-rulemaking. Additionally, FERC provides the resource *Energy Primer: A Handbook of Energy Market Basics* at www.ferc.gov/media/energy-primer-handbook-energy-market-basics.

[2] See "FERC Affirms Generator Interconnection Rule, Acts on Compliance Filings" at www.ferc.gov/news-events/news/ferc-affirms-generator-interconnection-rule-acts-compliance-filings.

[3] Transmission providers with wholesale distribution access tariffs (WDAT) will be allowed to submit their compliance filings within 90 calendar days of the date on which their relevant RTO or ISO submits its compliance filing.

[4] For more information on FERC's rulemaking process, see OPP's explainer at www.ferc.gov/OPP/rulemaking-explainer.

[5] See paragraph 9 of Order No. 2023 for a summary of how certain reforms adopted in the final rule deviate from the reforms proposed in the NOPR.

[6] See Order No. 2023 for summaries of comments that were submitted in Docket No. RM22-14-000.

[7] Joseph Rand et al., Lawrence Berkeley National Laboratory, Queued Up: Characteristics of Power Plants Seeking Transmission Interconnection, at 7-8 (Apr.2023) (Queued Up 2023), https://emp.lbl.gov/sites/default/files/queued_up_2022_04-06-2023.pdf.

[8] See Appendix B to this final rule, which provides an overview of recent data based on reporting by transmission providers in compliance with Order No. 845.

[9] See paragraphs 38-40 of Order No. 2023 for additional information and estimates on interconnection queue processing times.

[10] See paragraphs 10 and 1764 of Order No. 2023.

[11] Some transmission providers have already received FERC's approval to move to a cluster study process approach prior to the issuance of the final rule in an effort to manage interconnection queues. While recognizing these initiatives, the final rule still requires that all transmission providers submit compliance filings that justify how their intended practices meet the standards outlined in the final rule.

[12] For more information on alternative transmission technologies, please consider this U.S. Department of Energy (DOE) report, available here: <https://www.energy.gov/sites/prod/files/2021/02/f82/Advanced%20Transmission%20Technologies%20Report%20-%20final%20as%20of%2012.3%20-%20FOR%20PUBLIC.pdf>.

Quick Links

- [Explainer on the Interconnection Final Rule Requests for Rehearing and Clarification \(FERC Order No. 2023-A\)](#)
- [Explainer on the Interconnection Notice of Proposed Rulemaking \(NOPR\)](#)

Contact Information

Office of Public Participation (OPP)

Telephone: [202-502-6595](tel:202-502-6595)

Email: OPP@ferc.gov

News Media

Email: MediaDL@ferc.gov

This page was last updated on January 23, 2025